DEPARTMENT OF ECONOMICS
M.A. ECONOMICS SYLLABUS

Course No.	Paper Code	Name of the Course	Credit	Remarks
Semester I				
PAPER I	ECO-111	Macro- Economics	04	Core Course
PAPER II	ECO-112	Introduction to International Economics	04	Core Course
PAPER -III	ECO-113	Elementary Econometrics	04	Core Course
PAPER -IV	ECO-114	Economics of Agriculture	04	Value Added Course
PAPER -V	ECO-115	Introduction to Environmental Economics- I	04	Core Course
PAPER- VI-A	ECO-116A	Economy of Uttar Pradesh	04	Interdepartmental course
PAPER -VI-B	ECO-116B	Demography and Population Studies		
PAPER - VII	ECO-117	Research Methodology	04	Research Introduction
Semester Total			28	
Semester II				
PAPER I	ECO-121	Advanced Macroeconomics	04	Core Course
PAPER II	ECO-122	Advanced International Economics	04	Core Course
PAPER -III	ECO-123	Econometrics Theory & Application	04	Core Course
PAPER -IV	ECO-124	Environmental Economics- II	04	Core Course
PAPER -V-A	ECO-125A	Population theory and demographic analysis	04	Elective
PAPER -V-B	ECO-125B	Labour Economics		
PAPER -VI	ECO-126	Summer Internship / Field Work	04	Internship/ / Field Work
PAPER - VII	ECO-127	Computer Application in Economics	00	VAC (Non Credited)
Semester Total			24	

Semester III				
PAPER I	**ECO-131**	**Micro Economic Analysis – I**	**04**	**Core Course**
PAPER II	**ECO-132**	**Development Economics – I**	**04**	**Core Course**
PAPER -III	**ECO-133**	**Economics of Fiscal theory and public choice**	**04**	**Core Course**
PAPER- IV	**ECO-134**	**Advanced Indian Economy- I**	**04**	**Core course**
PAPER -V-A	**ECO-135A**	**Financial Institutions and Markets- I**	**04**	**Elective**
PAPER -V-B	**ECO-135B**	**Mathematics for Economic Analysis - I**		
PAPER -VI	**ECO-136**	**Project Dissertation**	**04**	**Preliminary**
Semester Total			**24**	
Semester IV				
PAPER I	**ECO0-141**	**Micro Economic Analysis- II**	**04**	**Core Course**
PAPER II	**ECO-142**	**Development Economics – II**	**04**	**Core Course**
PAPER III	**ECO-143**	**Government Budgeting and Fiscal Choice**	**04**	**Core Course**
PAPER -IV	**ECO-144**	**Advanced Indian Economy -II**	**04**	**Core Course**
PAPER -V-A	**ECO-145A**	**Financial Institutions and Markets- II**	**04**	**Elective Course**
PAPER -V-B	**ECO-145B**	**Mathematics for Economic Analysis II**		
PAPER -VI	**ECO-146**	**Master Dissertation**	**04**	**Master Thesis**
Semester Total			**24**	
GRAND TOTAL			**100**	

EC: Economics (Subject code)
ECIRA: Intradepartmental course
ECCC: Core course
EC-EL: Elective
ECIER: Interdepartmental Course
ECVC: Value added Course (Credit)
ECVNC: Value added Course (Non- Credited)
ECMT: Master Thesis

CURRICULUM & PROGRAMME STRUCTURE

M.A ECONOMICS

Purpose of the Programme: The programme aims to inculcate economic thinking among the students in economic decision making by comprehending economic theory and analytical view point in the students about the economic behavior of people. It develops socially responsible, ethically aware citizens and competent economic analysts who are ethically and socially responsible professionals.

(As per Bloom's Taxonomy)

PO1- Apply theoretical knowledge of micro Economics in real world problem solving. The students will be enhanced with skills like, research, statistics, mathematics and will be able to have employability in these areas. These skills will develop scientific temper and problem solving attitude.
PO2- To prepare the students to successfully compete for employment in Economics, industries, research methods, data analysis, etc. The course outcome of the above syllabus is to provide students with a comprehensive understanding of Statistics, including its definition, scope, importance, and limitations. Students will also learn about the various types of statistical data and the methods of collection and tabulation of data, as well as the various diagrammatic and graphic presentations.
PO3- To develop strong competencies in undergraduate level students in broad fields of Economics and its applications in an interactive environment. It will result in a comprehensive understanding of Indian Economy, govt policies and programs and how the implementation of these programmes benefit society.
PO4- To develop strong student skills in simulation, data analysis, and interpretation. The students will learn about progressions, coordinate geometry, integration, differentiation, etc. Students will learn about the economic application of these quantitative techniques, these skills will provide ample job opportunities for students.
PO5- To prepare the students to successfully compete for employment in Economics, industries, research methods, data analysis, etc. The course outcome of the above syllabus is to provide students with a comprehensive understanding of Statistics, including its definition, scope, importance, and limitations. Students will also learn about the various types of statistical data and the methods of collection and tabulation of data, as well as the various diagrammatic and graphic presentations.
PO6- It will result in a comprehensive understanding of Indian Economy, govt policies and programs and how the implementation of these programmes benefit society. The course will result in a comprehensive understanding of the Indian Economy and the economy of Uttar Pradesh. The course will promote critical thinking about various economic policies like five year plans.

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.Ă ECONOMICS

SEMESTER –I **PAPER CODE: ECO-111**

PAPER I: MACRO ECONOMICS

L	T	P	C
3	1	0	4

Course Objectives:

CO1: To develop an advanced theoretical understanding of basic macroeconomic variables

C02: To make learners capable of evaluating the impact of money demand on money supply and inflation

C03: To make learners understand the difference between neo classical and keynesian solutions to real world economic problems

C04: To enable understanding of the determinants of money supply and factors that affect money supply in india.

Learning Outcome:

After completion of the course students will be able to understand the basic concepts of Macroeconomics and their measurement. They will also be able to understand the various theories and approaches to macroeconomics pertaining to demand and supply of money.

UNIT – I

Introduction to Macro Economics and Macro Economic Measurement: Meaning and some preliminary concepts involved in Macroeconomic, Development of Macro-Economics : An overview, Macro-economic issues and their measurement : inflation, unemployment, output and growth, separating growth from cycles, Detrending a GDP time series, Macro-economic framework : An overview of the circular flow of income. Classical and Keynes theory of employment.

UNIT – II

Neo-classical and Keynesian Approaches to Macroeconomics : One sector Neoclassical model, Simple Keynesian model, economic implication of these models and essential differences between the two. Two sector IS-LM model : Both Neo-classical and Keynesian versions, Role and relative effectiveness of fiscal and monetary policies.

UNIT – III

Demand for Money : Classical approach to demand for money-quantity theory approach, Fisher's equation, Cambridge quantity theory, Keynes's liquidity preference approach, transaction, precautionary and speculative demand for money-aggregate demand for money; Derivation of LM curve.Post-Keynesian approaches to demand for money-Patinkin and the Real Balance Effect, Approaches of Baumol and Tobin; Friedman and the modern quantity theory; crisis in Kenynesian economics and revival of monetarism.

UNIT – IV

Supply of Money : A behavioral model of money supply determination, A demand determined money supply process, High powered money and money multiplier. RBI approach to money supply, Control of money supply with special reference to India.

Books recommended:-

Principles of Macro economics: Sampat Mukherjee and Amitav Ghosh. NCBA

Essentials of Micro and Macro economics: Debes Mukherjee New Central Book Agency.

Lectures on Macroeconomics: Oliver Jean Blanchard & Stanley Fischer. Prentice Hall India

PO-CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-112**

PAPER-II: INTRODUCTION TO INTERNATIONAL ECONOMICS

L	T	P	C
3	1	0	4

Course Objectives:

CO1: This course will enable students to understand the various theories and concepts of international trade.

CO2: This course provide a deep understanding of the broad principles and theories which tend to govern the free flow of international trade.

CO3: This course has become even more relevant with rising globalisation and liberalisation both in north and south.

CO4: This course will help to develop understading flow of international trade, movement of capital and labour and recording of such movements as they are the major determinant of development and change at the global level.

Learning Outcome:

The students should be able to understand the factors affecting international trade and understand the structure of various international institutions that govern international trade among countries, understand theories of international trade and as well as protection policies too.

UNIT – I

Pure theory of International Trade, Evaluation of the Classical Theory & Trade, Opportunity Cost approach to the theory of I.N.T., Terms of Trade, Concepts Measurements and Trends in Terms of Trade. Mill's Reciprocal Demand Analysis.

UNIT – II

Trade Policy; Free Trade-Evolution and Breakdown of free trade. Arguments for and against the Policy of Protection. Economic and Non-economic arguments for Tariff.

UNIT – III

Theories and Forms of Economic Integration. Custom Union: Production and Consumption effects of Customs Union-Static and Dynamic framework.

UNIT – IV

The Balance Payments Account and the foreign Exchange Market. Alternative Mechanisms of Adjustment in Balance of Payments.

Books Recommended:-

International Economics: Theory and Practice Paul R Krugman & Maurice Obstfeld Pearson.

World Trade and Payments:An Introduction. Richard E. Caves; Jeffrey A. Frankel & Ronald W. Jones .Pearson

International Economics: M. Maria John Kennedy. Prentice Hall India

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-113**

PAPER III: ELEMENTARY ECONOMETRICS

L	T	P	C
3	1	0	4

Course Objectives:
CO1: The course provide analytical approach to the learners helpful in drawing statistical inferences.
C02: To develop the understanding of basic econometrics and research ability.
C03: To impart knowledge of econometric tools and techniques, helpful in higher studies.
C04: The course aims to teach the applications of econometric methods in the analysis of production and consumption.
Learning Outcome : With the end of the course, students are successfully expected to understand the basic econometrics techniques and become prepared of learning advanced econometrics.

UNIT – I

Probability: Probability of an event, Theorems of total and compound probability, Baye's theorem, Mathematical expectations. Statistical Inference : Sampling-Random and Non-random sampling. Simple random, concept of an estimator and its sampling distribution, Desirable properties of an estimator.

UNIT – II

Definition and scope of Econometrics, Basic Linear Regression Model, Correlation and regression, The two variable linear model, The ordinary least squares methods (OLS), Assumptions, Gauss-Markov theorem, Significance tests of parameters, R^2 and adjusted R^2 . Properties of OLS estimators.

UNIT – III

Autocorrelation, Nature of the problem, Consequences of using OLS in the presence of Autocorrelation, Detecting Auto-correlation-Graphical Method, Durbin-Watson 'd- statistic', Remedial measures, Heteroscedasticity- meaning, graphical presentation of the presence of Heteroscedasticity. Consequences of Heteroscedasticity, Detection and remedial measures of the problem.

UNIT – IV

Application of Econometric Methods: Production Function, Micro and Macro Production Function, Properties and estimation of the Production Function, the Cobb-Douglas and CES Production Functions, Trans log and other forms of Production Function. Demand Analysis, Basic theory of Consumer Demand, Specification of the Demand Function, Price and Income elasticities of Demand, Engle's Law, Demand Equations-Linear, Semi- log and Log-linear forms, Systems of Demand Equation-Linear expenditure system, Aggregation and Data Problems in Estimation.

Books Recommended

Introductory Econometrics : A Modern Approach. Jeffrey Wooldridge
Introduction to Econometrics: Mark Watson & James H. Stock
Introduction to Econometrics: Christopher Dougherty.
Introduction to Econometrics: Johnston Macmillan Publishing House.

PO-CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	0	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-114**

PAPER- IV - ECONOMICS OF AGRICULTURE

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The programme aims to strengthen institutional mechanisms such as banks, insurance companies, and agricultural extension services to provide farmers with better access to credit, insurance, and information.

CO2: To provide detailed understanding regarding the Agriculture and Economic Development and various interaction forces that help in agricultural development.

CO3: To familiarize students with policy issues that are relevant to agricultural economics and enable them to analyze these issues.

CO4: To impart working knowledge of various agricultural finance institutions and how these help in economic growth.

Learning Outcome:

After completing the paper, the students are expected to understand and estimate the impact of investment and various other institutional reforms on agricultural production. The students are expected to differentiate between different types of agricultural growth, types of investment in agriculture.

UNIT – I

Agriculture and Economic Development : Interrelationship between Agriculture and Industry, Model of interaction between agriculture and rest of the economy. Allocation of investment between agriculture and industry in India during different plan periods. Trend in sectoral distribution of India's GDP and employment elasticity. Empirical generalization about the role of agriculture.

UNIT – II

Institutional Development for Agricultural Growth : Asset distribution of rural households in India over time, Indebtedness of rural households, non-institutional and institutional sources of credit, role of corporations, Schedule Commercial Banks, Regional Rural Banks and NABARD, Agricultural insurance.

UNIT – III

Infrastructural Development for Agricultural Growth: Tenancy in agriculture-incidence, effects and reforms, Land reforms in India, Irrigation development, Availability and use of HYV seeds and fertilizers in India.

UNIT – IV

Growth and composition of output in agricultural and allied activities in India : trends in overall agricultural growth as well as of circle, Pulses, Food and Non-food item. Demand and supply of agricultural commodities including plantation and Horticultural Produce. Demand, availability and Future prospects of livestock-, Poultry and Fishery in India.

Books Recommended:-

Principles of Agricultural Economics: Andrew Barkley & Paul W. Barkley

Agricultural Economics and Agribusiness: Gail Cramer

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-115**

PAPER – V : INTRODUCTION TO ENVIRONMENTAL ECONOMICS- I

L	T	P	C
3	1	0	4

Course Objectives:

CO1- To understand the basic concepts, scope and methodologies of environmental economics, institutions and environmental governance.

CO2- To analyze the relationship between ecology, environment and ecosystem, and explore the conflicting paradigms in economics regarding the balance between economic growth and environmental protection.

CO3- To examine the concept of sustainable development, and evaluate the indicators and accounting methods used to measure and monitor sustainable development.

CO4- To analyze the relationship between energy, infrastructure, trade, agriculture, industry, services sectors and the environment, and evaluate the economics of pollution and waste management.

Learning Outcome:

After completion of the course the student should be able to understand and solve basic questions about the methodological issues in environmental economics, sustainable development and many other factors which study interaction between economic development and environment.

UNIT – I

Introduction, Nature Scope and Methodological Issues in Environmental Economics, Institutions and Environmental Governance, Relation between development and environmental stress; Environmental Kuznet's Curve, Impact of Human Activity on Environment.

UNIT – II

Ecology, Environment and Ecosystem, Ecology and Economics, Economy-Ecology- Environment Interaction; Conflicting Paradigms, The Material Balance Principle: Ecological Concern in the evolution of Economic thought. Entropy Law, Economics of Exhaustible Resources: Hotelling's rule; Solow-Harwick's Rule; Market structure and optimal extraction policy; Uncertainty and the rate of resources extraction; Resource scarcity. Economics of Renewable Resources; Economic Models of forestry and fisheries; Extinction of species; Economics of Biodiversity.

UNIT – III

Concept of Sustainable Development, Sustainable Economic Growth and Development: An Economist's perspective. Rules of Sustainable Development, Indicators of Sustainable Development, Measuring Sustainable Development., Sustainable Accounting, Political Economy and Sustainable

Development, Participatory Institutions and Environmental Governance. Environmental Accounting.

UNIT – IV

Energy & Environment, Infrastructure and Environment, Trade and Environment, Agriculture and Environment, Industry and Environment, Services Sector and Environment, Economics of Pollution (air, Water, Solid Waste, Noise, Land degradation), Stationary and Mobile Sources of Pollution, Pollution Management, Waste Management Causes and effects of Environmental Degradation; Urban and Rural Environmental problems.

Books recommended:-

Environmental Economics: Barryfield & Marthafield. Tata McGraw Hill.

Environmental Economics: M.L.Jhingan Vrinda Publication

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	2	1	2	2
CO2	1	2	2	1	1	1
CO3	1	3	3	1	1	2
CO4	3	2	1	1	2	1

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-116A**

PAPER–VI A: ECONOMY OF UTTAR PRADESH

L	T	P	C
3	1	0	4

Course Objectives:

CO1:This paper aims to introduce students with the basic economic features of Uttar Pradesh economy.

CO2: The paper aims to understand the growth pattern, sectoral growth and development in Uttar Pradesh.

CO3: This course is designed to enhance the advanced understanding and knowledge of agricultural and rural development in Uttar Pradesh.

CO4: It will help in developing the knowledge about major industries of Uttar Pradesh in relevance to their working, development, policies and linkages with other sector.

Learning Outcome: By the end of the course, the learners will be able to get an macro and microscopic overview of the economy of Uttar Pradesh. Students will enlightened with the polices, grwoth pattern, sectoral developments, agricultural and rural development of Uttar Pradesh's economy.

UNIT I

Introduction Overview of Uttar Pradesh Economy Nature, Features, Demographic Profile, Status of Natural Resources, Major Factors affecting growth and development in Uttar Pradesh: economic and non-economic factors in economic development of Uttar Pradesh.

UNIT II

Growth Pattern of Uttar Pradesh Economy, Sectoral growth pattern of in Uttar Pradesh; Economic growth in Uttar Pradesh and Indian economy: A comparison; Infrastructural development of Uttar Pradesh.

UNIT III

Agricultural and Rural Development in Uttar Pradesh, Pattern of land-holding and irrigation; production and productivity in agriculture, Farm mechanization, Crop diversification, agricultural credit, Agricultural policy and strategies in Uttar Pradesh, Rural Development – Problems and Policies.

UNIT IV

Industry and Service Sectors Industrial and service sector development in UP Major industries in Uttar Pradesh, Pattern of Industrial Development in Uttar Pradesh, Industrial Policy in Uttar Pradesh, Growth pattern of Services sector and Its linkages of other sectors of UP Economy.

Suggested Readings:

1. Publications of the Government of Uttar Pradesh.
2. Uttar Pradesh Human Development Report, UNDP, 2008.
3. Uttar Pradesh State Development Report, Volume I & II, State Plan Division,
4. Planning Commission, Government of India, 2014
5. . M K Agarwal, Uttar Pradesh ka Arthik Vikas.

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW
(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-116B**

PAPER–VI B : Demography and Population Studies

L	T	P	C
3	1	0	4

Course Objective:
CO1: To enable learners appreciate the basic demographic concepts and its scopes.
CO2: To make learners understand the measuring of demographic processes
CO3: To enable learners appreciate the population policies and its impact on society
CO4: To enable learners comprehend different population health issues and apply the tool and technique to design and conduct demographic survey
Learning Outcome: By the end of this course, the student will able to learn some basic techniques and concepts in population sciences, Identify appropriate sources of data, Perform basic demographic analysis, Compare population and projections, interpret data using demographic methods.

UNIT– I

Basic Concepts and definition of demography, relationship between demography and anthropology & public health. Concepts of Population Growth and Projection, Demographic Transition, Demographic Dividend or Disaster. Importance of population studies and their implications & rapid population growth.

UNIT– II

Demography and health: Measures of demography – CBR, TFR, IMR, MMR, Life Expectancy, etc. Ageing; demographic composition, Vulnerable population- mother and child, Low SES –poverty, migrant, tribal, victims of war, refugees, disasters, street children, gender disparity. Demographic theories - Malthusian Theory, Optimum population Theory, Demographic Transition theory and population explosion, its impact on health etc. Various types of sampling design and their relevance in demographic studies. Sources of demographics data**:** census, CRS, SRS, national samples survey organization, model registration system.

UNIT– III

Factors affecting population Population structure: age and sex composition. Urbanization- Methods of measuring the size of urban population factors responsible for urbanization. Urbanization in India. Urban Agglomeration.

UNIT– IV

The basic components of population change – fertility, mortality, migration (their biological, social, ecological and economic factors influencing these components). Morbidity and its relationship with fertility and mortality. Reproductive health care in Indian population with reference to MCH, STDS, AIDS, Abortion, etc.

Suggested Readings

1. Bouge Donald: Principles of Demography, John Wiley & Sons, New York 22
2. Asha A Bhende & Thara Kanitkar: Principles of population studies, Himalaya Publishing House.
3. Park K: Text book of preventive and social medicine, M/s Banarasidas, Jabalpur
4. Cox P.R: Demography, England. Cambridge University Press; 5 edition (1976)

PO-CO MAPPING:

	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**
CLO1	3	2	3	3	3	3
CLO2	3	3	2	3	3	3
CLO3	3	3	3	2	3	3
CLO4	3	3	3	3	2	3

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –I **PAPER CODE: ECO-117**

PAPER VII: RESEARCH METHODOLOGY

L	T	P	C
2	1	1	4

Course Objectives:

CO1: The course aims to impart knowledge on defining and formulating the research problem, selecting the problem, necessity of defining the problem,

CO2: It aims to develop undestanding of various methods and techniques of data collection

CO3: It aims to enhance learning of hypothesis formulation and testing

CO4: It aims to develop research paper writing skills and techniques

Learning Outcome: After the completing of the course the students are expected to understand the basics of research and should be able to execute minor research works of primary and secondary data.

UNIT I

Introduction to RM; Meaning, Objectives, Categories of research, Features of good research studies, types of research studies, Scientific and non scientific methods, Research methods & research methodology.Process of research., Research design, Types of research design,

UNIT II

Methods and techniques of data collection; Types of data collection. Methods used for collection of different data types. Sampling and sampling distribution: importance of sampling, methods of sampling, sampling errors, Attitude measurement and Scales, Introduction to attitude- various methods to measure attitude

UNIT III

Types of Hypotheses; Formulation of Hypothesis; criteria of a good hypothesis, types of research hypothesis, advantages of Hypothesis. Hypothesis Testing Procedure, Type I and II Error, Chi-square test (Theoretical concepts & very simple numerical problems only)

UNIT IV

Research paper wiring and presentation

Books Recommended

C.R Kothari : Research Methodology

Dr. Kumar : Research Methodology

PO-CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-121**

PAPER I: ADVANCED MACROECONOMICS

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The students should be able to solve basic consumption functions, understand the shape and interaction of demand and supply curve

CO2: The students should also be able to derive the relationship between unemployment and inflation in different settings.

CO3: The course will equip students with the knowledge and skills to analyze macroeconomic policies and make informed decisions. They will be able to evaluate the impact of monetary and fiscal policies on inflation, employment, and income distribution,

CO4: The students will be able to recommend appropriate policy measures to achieve macroeconomic stability and economic growth

Learning Outcome:

By the end of this course, the student will able to understand the various advanced concepts of Macroeconomics. The will also be able to understand the various approaches pertaining to different school of thoughts such as neo-classical, keynesian and post keynesian.

UNIT I

Consumption Function: Short Run and Long Run, Consumption Functions, Alternative theories of consumption function: Absolute, Relative, Permanent and Life cycle hypotheses.

Investment function : Investment-interest rate relationship, Investment demand function, Investment income relationship. Accelerator and Investment behavior, Multiplies- Accelerator Models.

UNIT II

Neo-classical and keynesian approaches to the labour market equilibrium. Three sector macro model and derivation of aggregate demand and aggregate supply curves in the Neo-Classical and Keynesian cases. Interaction between aggregate demand and aggregate supply curves. Application and limitations of the Neo-Classical and Keynesian Models, Neo Classical-Keynesian synthesis and its policy implications. Walras general equilibrium analysis.

UNIT III

Post Keynesian Economics : Rejection of general equilibrium analysis and main ideas of post Keynesian Economists; Neo-Keynesian disequilibrium approach (Quantity- Constrained Models) and its policy implications. Early Phillips curve and Relevance of Keynesian Economics, Friedman, Phelps hypotheses and role of expectations; Expectation augmented Phillips Curve and policy relevance.

UNIT IV

Phillips Curve and aggregate Supply Curve, Alternative theories (Neo Classical and Neo Keynesian approaches) regarding aggregate supply curve, classical and Keynesian theories in a dynamic setting, Neo Keynesian wage equation and synthesis of the dynamic classical and Keynesian approaches. Expectation of inflation in dynamic setting, Practical Relevance of dynamic approach.

Books recommended:-

Principles of Macro-economics: Sampat Mukherjee and Amitav Ghosh. NCBA

Essentials of Micro and Macro economics: Debes Mukherjee New Central Book Agency.

Lectures on Macroeconomics: Oliver Jean Blanchard & Stanley Fischer. Prentice Hall India.

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-122**

PAPER II: ADVANCED INTERNATIONAL ECONOMICS

L	T	P	C
3	1	0	4

Course Objectives:

CO1- To make learners understand the Heckscher-Ohlin theory of trade, including the assumptions, predictions, and empirical evidence supporting the theory.

CO2- To make learners analyze the history and evolution of the General Agreement on Tariffs and Trade (GATT) and the World Trade Organization (WTO), including the major rounds of multilateral trade negotiations and the key principles of the WTO's legal framework.

CO3- To make learners evaluate the effects of trade restrictions on welfare, employment, and the distribution of income, using the general equilibrium model. Students will explore different types of trade barriers, including tariffs, quotas, and non-tariff measures, and analyze their impact on trade flows, prices, and economic welfare.

CO4- To learn about foreign exchange reserves, their role in managing exchange rate volatility, and the challenges facing countries in maintaining adequate reserves.

Learning Outcome:

The students should be able to understand the factors affecting international trade and understand the structure of various international institutions that govern international trade among countries.

UNIT I

Factor Proportion Theory: H-O Theorem, Factor Price equalization, Effects of Growth on International Trade, Effects of Factor Endowment, Change in Technology, Change in demand Pattern, Gains from Trade.

UNIT II

Trade Policy for Developing Countries, GATT, WTO; Agreements on Agriculture aoa), Textile WTO and India; Doha, Cancun, Hong-Kong, North-South Trade Dialogue, UNCTAD, South-South Cooperation.

UNIT III

General Equilibrium in International Trade, Quantitative Restriction: Quota and Tariff.

UNIT IV

Foreign Exchange Reserve; Magnitude Components, Convertibility of Rupee in Current Account and Possibility in Capital Account. East-Asian Crisis.

Books Recommended:-

International Economics: Theory and Practice Paul R Krugman & Maurice Obstfeld Pearson.

World Trade and Payments:An Introduction. Richard E. Caves; Jeffrey A. Frankel & Ronald W. Jones .Pearson

International Economics: M. Maria John Kennedy. Prentice Hall India.

International Economics: Bo Sodersten and Jeffory Reed

PO- CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	2	1	1	2
CO2	1	2	2	1	1	1
CO3	1	3	3	1	1	1
CO4	3	2	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-123**

PAPER III: ECONOMETRIC THEORY AND APPLICATION

L	T	P	C
2	1	1	4

Course Objectives:

CO1: To provide analytical approach to the learners, helpful in drawing statistical inferences and empirical studies.

C02: To develop the conceptual understanding of advanced econometric methods and models.

C03: To impart knowledge of advanced econometric tools and techniques, helpful in advanced studies.

C04: The course aims to provide knowledge about applications of econometric methods in economic theories.

Learning Outcome: The ability to conduct empirical analyses and data analytics are increasingly valued in the job market. This course will enable students to understand why and how questions are to be framed and answered. It will also equip them to learn more advanced topics on their own.

UNIT I

Probability Distributions : Binomial, Poisson, Normal Distributions, their properties and uses. Statistical Inference: Testing of Hypotheses, Formulation of statistical hypothesis-Null and alternative, Type I and II errors, Goodness of fit, Confidence intervals and level of significance, Hypothesis testing based on Z, t, χ^2 (Chi-square) and F test.

UNIT II

Multiple regression with two explanatory variables, The coefficient of Multiple Determination, Test of overall significance of the Regression, Partial correlation coefficients. General Linear Model-Estimation and properties of estimators.

UNIT III

Problems in Regression Analysis: Nature, Tests, Consequences and remedies of Problem of Multicollinearity, Problems of specification error, Errors of Measurement.

UNIT IV

Applications of Econometric Methods: Estimation of Consumption & Investment functions, Theories of Consumption :- Permanent Income Hypothesis, Absolute Income Hypothesis, Life Cycle Hypothesis Relative Income Hypothesis; Investment Function- Accelerator theory of Investment-Crude and Flexible Accelerator Models, Neo-Classical theory of Investment-Jorgenson's Model.

Books Recommended
Introductory Econometrics : A Modern Approach. Jeffrey Wooldridge
Introduction to Econometrics: Mark Watson & James H. Stock
Introduction to Econometrics: Christopher Dougherty.
Introduction to Econometrics: Johnston Macmillan Publishing House.

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-124**

PAPER IV: ENVIRONMENTAL ECONOMICS –II

L	T	P	C
3	1	0	4

Course Objectives:

CO1- The course aims to enhance the understanding of the concept of public goods and externalities, and how they relate to environmental issues.

CO2- It aims to develop learning about environmental valuation techniques, such as contingent valuation and hedonic pricing, and how they can be used to estimate the economic value of environmental goods and services.

CO3- It aims to enhance knowledge on different types of environmental policy instruments, including command-and-control regulations, market-based mechanisms (such as emissions trading and carbon taxes), and voluntary agreements.

CO4- It aims to impart understaing on examining the links between population growth and environmental degradation, including the concept of carrying capacity and the challenges of sustainable development.

Learning Outcome:

The students should be able to understand the conditions of welfare caused by the environment and public goods, while at the same time understanding the impact of externalities and various methods of evaluation of the cost of public goods payment mechanism.

UNIT I

The theory of Externality and Public Goods: Concepts, Market failure, Pigouvian Solution, Buchanan's Theory, Coase's theorem and its critique, Pigouvian vs. Coasian solution; Detrimental externality and non-convexities in the production set; Property rights; Collective action.

UNIT II

Measurement of Environmental Values: Use values; Option value and non-use values. Techniques of Valuation: Market and Non- Market Valuation: Physical linkage methods; Abatement Cost Methods; Behavior Linkage Method-Revealed and stated preference; Social Cost benefit Analysis; Environmental impact assessment, Hedonic property values and household production models (Travel cost method and household health production function). Methods based response to hypothetical Markets, Contingent Valuation Methods.

UNIT III

Environmental and Natural resource policy and strategies, Evolution of Environmental Policy, Economics of Environmental Policy. Protection, Conservation. Policy Instruments; Pigouvian Tax and Subsidies, Tradable permits and fees, Price vs. Quantity instruments, Mixed instruments,

Monitoring and Enforcement Issues, India's Environmental Policy and Performance, International Environmental Policy, Environmental Planning and Management, Macroeconomic Policies and Environment , People's Participation.

UNIT IV

Population and Environment, Poverty and Environment, Trade and environment in the WTO regime, Eco-labeling, Economics of climate change, Causes and Consequences of Ozone depletion and climate change. Rio and other then Rio Conference, Protocols and Treaties relating to climate change and Environmental degradation, Ozone depletion and Biodiversity.

Books recommended:-

Environmental Economics: Barryfield & Marthafield. Tata McGraw Hill.

Environmental Economics: M.L.Jhingan Vrinda Publication

PO-CO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1	1	1	1	1	1
CO2	1	2	2	1	2	2
CO3	2	3	3	1	2	1
CO4	2	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II PAPER CODE: ECO-125A

PAPER–V A: POPULATION THEORY AND DEMOGRAPHIC ANALYSIS

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The course aims to make the students aware of basic theories of population and demography.

C02: To develop the understanding in various measures of demographic analysis.

C03: To impart knowledge about the importance of population in economic development through various models of population and development.

C04: The provide the knowledge about the sources of data of population and demographic changes helpful in conducting research in future.

Learning Outcome:

By the end of this course, the student will able to learn some basic techniques and concepts in population sciences, Identify appropriate sources of data, Perform basic demographic analysis, Compare population and projections, interpret data using demographic methods.

UNIT I

Concept of Population and Demography; Biological Theories; Theories of Population: Malthusian, Post-Malthusian, Neo-Malthusian; Optimum theory; Modern theories and Recent Developments; Theory of Demographic Transition.

UNIT II

Measures of Demographic Analysis; Fertility, Mortality, Morbidity; Population Ageing; Migration.

UNIT III

Population and Economic Development; Models of Population and Economic Development; Demand for Children in Poor Countries; Demographic Variables and Income Inequality.

UNIT IV

Indian Population and Demographic Changes; Inter-State Variations; Population Policy; Sources of Data-Census, SRS, NFHS, RCH etc.

Books Recommended:-
Demography: M.L.Jhingan, B.K.Bhatt & J.N.Desai
Human Development and Economic Development: Ruddar Dutt
Population Growth and Economic Development in Low Income Countries: A.J. Coale & E.M. Hoover.

PO-CO MAPPING:

	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**
CLO1	3	2	3	3	3	3
CLO2	3	3	2	3	3	3
CLO3	3	3	3	2	3	3
CLO4	3	3	3	3	2	3

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-125B**

PAPER–V B: LABOUR ECONOMICS

L	T	P	C
3	1	0	4

Course Objectives:
CO1: To develop students' abilities in acquiring a better understanding of the functioning of labour markets and various issues.
CO2: To thoroughly understand the various wage determination theories and payment concepts.
CO3: To gain knowledge and insights on dynamics of Industrial Relations and Dispute settlement methods.
CO4: To enhance understanding of Labour welfare and social security measures in India and abroad.
Learning Outcomes: Acquaint students with the contribution of labour in the process of capitalist development. Exposing students to diverse labour relations in urban and rural settings under capitalism. To enable students to understand the crucial role of women in the process of production

UNIT I

Labour: Some key Issues, Labour in a Developing Economy, Structure and Peculiarities of Labour Market, Indian Labour; Characteristics of Indian Labour, Recruitment of Industrial Workers-Employment Exchange, National Employment Service in India, Rural Labour in India, Unemployment, Employment Programmes in India, Labour Productivity and Wage Relationship.

UNIT II

Wages Determination: Theories of Wage Determination, System of Wages Payment Various Concepts of Wages-fair.Trade Unionism- Theories, Functions, Trade Union Movement Abroad.

UNIT III

Industrial Relations; Industrial Disputes and Establishment of Industrial Peace, Methods of Settlement of Industrial Disputes; Collective Bargaining, Conciliation, Arbitration and Adjudication.

UNIT IV

Labour Welfare: Labour Welfare Measures in India, Social security- Social Security in India,

Social Security Abroad.

Books Recommended:-
Labour Economics: Stephen Smith.
Labour Economics: Paul l. Latreille & Peter j. Sloane.

PO-CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-126**

PAPER–VI: SUMMER INTERNSHIP / FIELD WORK

L	T	P	C
1	1	2	4

Course Overview:

The objective of this course is to develop the human and soft skills of the students of Master's Degree. The internship track is designed for students who want to engage in real economic analysis outside of the classroom. The 04 credit internship is designed to expose students to a variety of contexts in which economists work within the development sector, in higher education, in public policy and with government agencies.

General Outlines:

1. The course should make the students learn how to prepare project reports on the basis of Internship and the student must provide the certificate of successful completion of internship from the respective company/ institute.
2. The preparation of a project report on a given topic should be made referring to Internship experiences.
3. The project report based on internship and its presentation should be evaluated by a duly constituted Department level committee.
4. The Internship/ project work will be evaluated via Interview.

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –II **PAPER CODE: ECO-127**

PAPER–VII: COMPUTER APPLICATIONS IN ECONOMICS

L	T	P	C
1	1	2	4

Course Objectives:

CO1: To enhance practical knowledge on performing data processing techniques using MS Excel, including data entry, file handling and operations, data storage and retrieval, and graphical representation of data.

CO2: It aims to develop interest among students regarding use of computers in Economics. The usage of statistical tools, software and preparation of data files will make the student industry ready.

CO3: It aims to develop skills in summarizing and analyzing data using MS Excel, including descriptive statistics, comparison of means, correlation and regression analysis, and estimation of growth rates and trends.

CO4: It aims to impart knowledge and practical skills on SPSS usage in data analysis and create understanding on statistical package handling and command description.

Learning Outcomes:

After completing the course the students will be able to use statistical software applications such as SPSS for data analysis, including data entry, computing, graphing, and command descriptions for reports.

UNIT I

Computer Fundamentals Introduction to Computers; Classification of Digital Computer, Meaning and Concept; Hardware, Software and Types of Software; Operating System Meaning and Functions; Memory – RAM and ROM; Input-Output Devices; MS Office: MS-Word and MS PowerPoint

UNIT II

Data Processing Techniques using MS Excel Concept of Data – Record and File – Types of Data – Data Entry – File handling and Operations like opening, appending and cascading – closing and attribute controls – Data Storage and Retrieval, Graphical Representation of Data.

UNIT III

Application of MS-Excel Summarizing and analysis of data; Descriptive Statistics (Mean, median, mode, standard deviation, CV, skewness); Comparison of means; Correlation and

Regression analysis; Estimation of Growth Rates; Trends of Forecasting; Testing the significance of parameters.

UNIT III

Internet and E-Commerce E-Commerce: Meaning, E-Commerce, E-Business and E-Transaction; Introduction to Statistical Software and Estimation Basics of Data Analysis – Data Entry in SPSS – Computing with SPSS – Preparation of Graphs with SPSS – Distribution Functions and Density Functions – Statistical Package handling and command description for SPSS – Reports, Descriptive – Statistics, Compare Means, Time Series Analysis, Correlation and Regression Models.

Suggested Readings

Fundamentals of Computers: V. Rajaraman, Prentice Hall India Learning Private Limited; 5 edition (2010)

E-Commerce: Strategy, Technologies and Applications: David Whiteley, Tata McGraw Hill, New Delhi.

Computer Application to Business & E-Commerce: Sushil Goel, Natraj Publishing House, Karnal. **Theory and Problems of Data Processing:** M.M. Lipschultz, and S. Lipschultz, Schum's Outline Series, McGraw Hill, New York.

Computer Applications in Business: R. Parameswaranm., S. Chand and Company, New Delhi

PO-CO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**

0	No Correlation

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –III **PAPER CODE: ECO-131**

PAPER–I: Micro Economics Analysis – I

L	T	P	C
3	1	0	4

Course Objectives:

CO1: To develop an advanced theoretical understanding of consumer behavior and decision-making.

CO2: To make learners understand the various concepts related to production theories, technical progress and cost of production.

CO3: To develop understanding of various market forms, their structures, models and pricing in these makets

CO4: To introduce students to the concept of welfare economics and its related theorems.

Learning Outcome:

On successful completion of the course,a student will be able to develop as an understanding of the core microeconomic concepts that economists use to understand the process of decision-making by an economic agent.

UNIT – I

Consumer Behavior: Measurement of Utility, Cardinal vs Ordinal Approaches, Indirect Utility Function, Roy's Identity, Theory of Revealed Preference and logic of Ordering; Strong and weak Ordering and theory of Demand; Hicks Revision of Demand theory and characteristics approach to Demand.

UNIT – II

Theory of Production: Production Function of a Single and Multi Product Firm, Technological Progress and Production Function, Choice of Optimal Input Combinations. Cost Function: Meaning and Types of Cost, Traditional and Modern Theory of Cost, Shapes of Cost Curves, Analysis of Economies of Scale.

UNIT – III

Markets Form and Pricing of Product: Types of Market Situations, Relationship between Perfect and Monopoly Situation, Duopoly Market Situation, Cournot's, Stackelberg, Edgeworth and Bertrand Models, Monopolistic Competition: Characteristics, Price and Non-Price Competition.

UNIT – IV

Welfare Economics: Old Welfare Economics: Marshall and Pigou, Pareto and Welfare Economics: Criteria and Optimality, Compensation Principle: Hicks, Kaldor and Scitovsky, Social Welfare Function: Bergson and Samuelson, Arrow on Social Choice and Individual Values.

Books Recommended:-

Modern Microeconomics: A. Koutsoyiannis Palgrave MacMillan

Microeconomics: Robert Pindy Pearson

Principles of Microeconomics: N. Gregory Mankin

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –III **PAPER CODE: ECO-132**

PAPER II: DEVELOPMENT ECONOMICS – I

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The objective of the current course is to incline the mechanical understanding of the development process, which includes differencing between development and growth, causes of poverty, factors causing circular causation, social dualism.

CO2: The course will develop the understanding of different models of development which range from learning by doing to Kaldor models of growth.

CO3: The course will enhance the in depth understanding of various growth theories as well as growth models.

CO4: This course will enable knowledge about the various policy implications of development theories and impart knowledge about the wider perspectives of development including human development.

Learning Outcome

After completion of the course the students should be able to solve basic numerical questions about different growth models. The student should develop understanding of the models like Optimal growth theory-Ramsay, Von Neumann, Turnpike theorems, Golden Rule of Accumulation Endogenous growth and Intellectual capital.

UNIT – I

Meaning and Measurement of Economic Growth and Development, General theories of Growth and Development: Classical theory, Smith, Ricardo, Marxian theory of capitalist development, Rostow's stages of growth theory

UNIT – II

Modern Theories of Economic Growth: Conceptual framework and methodology of modern growth theories. Harrod-Donar growth model. Contemporary growth theories of neo-classical: Solow, Meade, Uzawa.

UNIT – III

Cambridge theories of growth, Joan Robinson, Kaldor, Pasinetti, the capital controversy. Money and economic growth- Tobin, Levhari, Patinkin and Johnson's models.

UNIT – IV

Development Policies and Perspectives- Role of economic and non-economic factors, sectoral Priorities and Development policies, role of the informal sector, environmental issues in development, Human Development- HDI, food security, education, health and nutrition, Gender and Development.

Books Recommended:-

- The Economics of Development and Planning: R.K.Lekhi & Joginder Singh .Kalyani Publishers
- Development Economics- A Critical Appraisal: Aparajita Mukherjee & Saumya
- Economic Development: C.P. Kindleberger
- Leading Issues in Economic Development: G.M. Meier

- **CO-PO MAPPING:**

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –III **PAPER CODE: ECO-133**

PAPER III: ECONOMICS OF FISCAL THEORY AND PUBLIC CHOICE

L	T	P	C
3	1	0	4

Course Objectives:

CO1: This course provides a foundation for the study of fiscal functions and theories of public expenditure.

CO2: It will help students understand the budgetary process and union -state financial relations.

CO3: It will develop the understanding of taxation policy and various forms of taxes in learners.

CO4: This Course will make learners able to learn about the pricing of public utilities and debt of government.

Learning outcome:

After completing the course the students should be able to understand the policy issues ranging from production sectoral preference to evolution of industrial developments in india.

UNIT – I

Overview of Fiscal Function: Allocation, Distribution and Stabilization, Non-Fiscal and Quasi-Fiscal Government Activities. The Pure Theory of Public Expenditure and Its Integration with Welfare Maximization, Models of Efficient Allocation, Merit Goods.

UNIT – II

The Budget Cycle and the Budgetary Process: Classification of Budgets and Budgetary Practices: Fiscal Federalism: Union-State Financial Relations in India: Finance commission.

UNIT – III

Structure of Government Income, Principles of Taxation, the Allocation Effects of Taxation. Taxation of Income (Individual) and Corporate; Wealth and Consumption: Taxation of Agricultural Incomes, Commodity Taxation on Production Base, Value Added Tax and Modified Value Added Tax.

UNIT – IV

Public Debt: Ownership and Consumption of Public Debt, Macro Economic Policy and Economic Effects of Public Debt, Management of Public Debt; Pricing of Public Utilities; Marginal Cost Pricing, Postal Services, Railways and Electricity.

Books Recommended:
Public Finance: Richard A. Musgrave
Public Finance: Musgrave & Musgrave
Public Finance: B.P.Tyagi

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –III **PAPER CODE: ECO-134**

PAPER IV: ADVANCED INDIAN ECONOMY-I

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The objective of the course on the Indian economy is to develop the understanding of the economic structure of India, which ranges from analysis of national income, pattern of distribution of consumption pattern, energy infrastructure transport and employment.

CO2: This course will enhance the knowledge of infrastructure of Indian economy and its various components including economic as well as social.

CO3: This course will provide deep understanding about the external sector policies and regimes. It delves into the foreign linkages and trade.

CO4: The student should develop insight into the economy of Uttar Pradesh, Five-year plans and social sector development of Uttar Pradesh.

Learning Outcome:

After the completion of course the students should develop understanding and answer basic questions regarding foreign trade, investment WTO and trade policy and reforms.

UNIT – I

National Income and Pattern of Consumption in Indian Economy, Saving, Investment and Capital Structure, Resource Mobilization, Other Major Macro Economic Issues.

UNIT – II

Infrastructure of Indian Economy; Economic Infrastructure: Energy, Transport and Communication, Banking, Finance, Science and Technology. Social Infrastructure: Education, Health, Social Welfare, Environment and Employment.

UNIT – III

External Sector of the Economy: Evolution of Trade Policy in India. Trade Policy Reforms, Globlisation, Foreign Direct Investment, Recent Development in Foreign Trade of India, Convertibility, WTO & India.

UNIT – IV

Recent Trends and Emerging Issues in the Economy of the U.P. Five year Plans of U.P., Agriculture, Industry and Social Sector Development.

Books Recommended:-

Indian Economy: Ruddar dutt & K.P.M. Sundaram. S. Chand

Indian Economy: A.N. Agarwal & M.K. Agarwal

Indian Economy: I.C. Dhingra

India after the Global Crisis: Shankar Acharya . Orient Blackswan

Indian Economy since Independence: Uma Kapila. Academic Foundation.

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER –III **PAPER CODE: ECO-135A**

PAPER V A: FINANCIAL INSTITUTIONS AND MARKETS I

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The course will help in identifying the functions of financial markets and institutions and examine their impact on the level of interest rates and interest differentials.

CO2: It will develop deep understanding about the structure, role and instruments of money markets as well as the pricing and valuation of money market instruments.

CO3: It will provide insights about the role and functions of commercial bank including the in depth of monetary policy.

CO4: It will introduce students to the concepts of fintech, blockchain, cryptocurrencies and sustainable finance investing.

Learning Outcome: After completing the course the students should be able to answer the basic questions regarding financial institutions, understand the working of the financial markets and the role of financial institutions in economic growth

UNIT – I

Financial markets, Types of Financial Market, characteristics of financial market, Role of financial institutions in the economy, Regulatory framework and financial market infrastructure.

UNIT – II

Introduction to the money market, structure, Role, characteristics, defects of money market. Instruments of Money Market: Treasury bills, call/notice money, Commercial paper, Certificate of Deposit, Repurchase agreements (Repos), Money market mutual funds, Pricing and valuation of money market instruments.

UNIT – III

Role and function of commercial bank, Risk management in banking, Basel III framework, Functions of central banks, Monetary policy tools and objectives,Impact of monetary policy on money markets, Role of commercial bank and central bank in Indian Money Market.

UNIT – IV

Fintech and its impact on financial institutions, Blockchain and cryptocurrency, Sustainable finance and socially responsible investing, Evolving trends in money markets and financial institutions.

Books Recommended:-
Indian Financial System: Bharti Pathak McGraw Hill Education
Financial Institutions and Markets: L.M.Bhole & Jitendra Mahakud
Financial Market and Institutions: Mishkin. Pearson

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW
(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – III Paper code: ECO-135B

PAPER V B-:Mathematics for Economic Analysis I

L	T	P	C
3	1	0	4

Course Objectives:
CO1: The course will help in learning the concept and properties of determinants and matrices.
CO2: It will develop deep understanding about the elements of trigonometry and plotting of curves.
CO3: It will develop leaarning about differential calculus and its various calculation as well as economic application.
CO4: It will develop leaarning about integral calculus and its various calculation as well as economic application.
Learning Outcome: After completing the course the students should be able to answer the basic questions regarding financial institutions, understand the working of the financial markets and the role of financial institutions in economic growth

UNIT I

Determinants, Properties of determinant, Minors and Cofactors, Jacobian and Hessian determinants and their use. Matrics : Definitions, Addition, Subtraction and Multiplication of matrices; Inverse matrices; Rank of matrix; Solving matrix equations with inverse; Eigen values and Eigen vectors.

UNIT II

Elements of Trigonometry and Coordinate Geometry; Trigonometric functions and identities. Two –dimensional coordinates Plotting of curves; Straight line; Parabola and Hyperbola.

UNIT III

Differential Calculus : The derivative. Higher order derivative; maximization and minimization of a function; Partial derivatives; Differentials, Total differentials, Rules of Differentials; Optimization of a function without and with constraints. Economic Application.

UNIT IV

Integral Calculus ; Indefinite and definite integration; Rules of integration; Integration by Substitution; Integration by parts. Economic Applications.

Books recommended:-

Mathematical Economics: Alfac. Chiang
Mathematics and Statistics for Economics: Mongia
Mathematical Methods and Models for Economists: Angel de la Fuente and Antonio Manresa
Mathematics for Economists: Simon and Blume
Advanced Microeconomic Theory: Geoffrey A. Jehle and Philip J. Ren

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – III **Paper code: ECO-136**

PAPER VI-: PROJECT DISSERTATION

L	T	P	C
1	1	2	4

Course Overview: The objective of this course is to develop the research skills of the students of Master's Degree. The project dissertaiondesigned for students who want to engage in real economic research issues and analysis which will be much more than classroom teaching. The students will learn about tools and techniques of research and its application in their own research work. The 04 credit project dissertation is designed to expose students to a variety of research problems in which researchers collect data and produce results on the basis of them which have high policy and practical implications.

General Outlines:

1. Each student should be assigned a project supervisor from the faculty members of the concerned center.
2. The topic of the project report should be decided by the supervisor taking into account the area of interest of the student. As far as possible the topic should address the current issues / developments in the field of Economics.
3. Standard norms for the preparation of a research project report may be followed – introduction, brief background, relevance of the topic, literature review, problem statement, objectives, hypotheses, sample design, data nature and sources, tools and techniques of data collection, data presentation, analysis, and interpretation, findings and conclusion.
4. The Dissertation outcomes may be reported in the following form:

Chapter-1: Introduction & Review of Literature;

Chapter-2: Research Methodology;

Chapter-3: Conceptual & Theoretical Description;

Chapter-4: Analysis and Interpretations;

Chapter-5: Conclusion, Suggestions & Scope for further research; References, Annexure, etc.

5. The Dissertation outcomes should be evaluated both in the form of reports and viva voce examination by a duly constituted committee by the appropriate authority.
6. This is to be evaluated as 'Satisfactory (S)' or 'Unsatisfactory (US)'

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV Paper code: ECO-141

PAPER I: MICRO ECONOMIC ANALYSIS – II

L	T	P	C
3	1	0	4

Course Objectives:

CO1: The objective of this course is to make students understand about the concept of market failure and its various forms.

CO2: It will develop deep understanding about the various approaches to output and price determination as well as various models in oligopolistic market.

CO3: It will make learners understand about the various concepts of games theory and its related strategies.

CO4: It will develop knowledge the various theories of distribution including theories of rent, profit, interest and wages.

Learning Outcome: The students should be able to solve basic utility functions, understand the price determination in different market structures and should be able to solve basic questions regarding theories of rent profit and wages

UNIT – I

Market Failure and Asymmetric Information: Quality Uncertainty, Market Signaling, Moral Hazard and Principal – Agent Problem. Measurement of Utility: Neuman- Morgenstern Utility Index.

UNIT – II

Market and Price Determination: Oligopoly; Non-Collusive and Collusive Oligopoly, Kinked Demand Analysis, Role of Cartel and Price Leadership Models. Merger and Acquisition, Types of Mergers and Economic Theories Regarding Mergers. Limit-Price Theorem: Role of Natural and Strategic Barriers in the Determination of Limit Price.

UNIT – III

Game Theory: Zero-Sum Game, Pure and Mixed Strategies under Zero-Sum Games; Variable Sum Game; Concept of Dominant and Dominated Strategies, Mixed Strategies under Variable Sum Game. Alternative Methods of Determination of Optimal Pay-Offs and Nash Equilibrium. Prisoners' Dilemma and Folk Theorem.

UNIT – IV

Theories of Distribution: Theories of Rent, Interest, Profit and Wages. General Equilibrium and Economic Efficiency, Efficiency and Attainment of General Equilibrium.

Books Recommended:-

Modern Microeconomics: A. Koutsoyiannis Palgrave MacMillan
Microeconomics: Robert Pindy Pearson

Principles of Microeconomics: N. Gregory Mankin

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV Paper code: ECO-142

PAPER II: Development Economics -II

L	T	P	C
3	1	0	4

Course Objective:
CO1: The objective of the current course is to incline the mechanical understanding of the development process, which includes differencing between development and growth, causes of poverty, factors causing circular causation, social dualism.
CO2: The course will develop the understanding of different models of development which range from learning by doing to Kaldor models of growth.
CO3: It will provide knowledge about the various growth theories and their related concepts such as golden rule of accumulation and endogeneous growth.
CO4: It will make learners understand about the various economic problems and issues and delves into the areas of international trade and international institutions.
Learning Outcome: After completion of the course the students should be able to solve basic numerical questions about different growth models. The student should develop understanding of the models like Optimal growth theory-Ramsay, Von Neumann, Turnpike theorems, Golden Rule of Accumulation Endogenous growth and Intellectual capital.

UNIT – I

Development Underdevelopment : An overview Meaning and Measurement of Economic growth and development, the characteristics and explanations of underdevelopment – Vicious circle of poverty, circular causation, Dualism-social, technological, financial, organizational, Model of dual economy, Lewis, Ranis-Fei, Jorgenson, Kelly et al, dependency theories of underdevelopment.

UNIT – II

Models of Technical Progress, embodied and disembodied technical progress, Neutral technical progress-Hicks, Harrod, Solow, Kaldor, Mirrlees technical progress function, Arrow's learning by doing approach to economic growth, total factor productivity and growth accounting.

UNIT – III

Optimal growth theory-Ramsay, Von Neumann, Turnpike theorems, Golden Rule of Accumulation Endogenous growth, Intellectual capital, role of learning, education and research, AK model, Explanations of cross country differentials in economic growth, Information Paradigm-Stiglitz.

UNIT – IV

International Trade, Aid, Finance in Development, Technology, Transfer and Multinational Corporations. Dual Gap Model, W.T.O., Regional Cooperation. Problems of Unemployment, inequality and poverty in developing economies. Economic development and institutions-markets and market failure, state and state failure, issues of good governance.

Books Recommended:-

The Economics of Development and Planning: R.K.Lekhi & Joginder Singh .Kalyani Publishers

Development Economics- A Critical Appraisal: Aparajita Mukherjee & Saumya Chakraborti.

Economic Development: C.P. Kindleberger

Leading Issues in Economic Development: G.M. Meier

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV **Paper code: ECO-143**

PAPER-III- GOVERNMENT BUDGETING AND FISCAL CHOICE

L	T	P	C
3	1	0	4

Course Objectives:

CO1:T The objective of the course is to develop understanding about fiscal decisiom making process and models of voting in the context of public policies.

CO2: It will make students aware of the functionality of the budgeting mechanism including budget cycle, practices as well as various types.

CO3: It will incline the learners towards in-depth evaluations techniques of fiscal policy and its applicability to less developed countries.

CO4: It will make learners understand about the cencept fiscal decentralization as well financial administration of local governments.

Learning Outcome: The students should be able to differentiate between different tools of fiscal policy which are used to achieve its objectives. From understanding of budgets, modern public choice theory to fiscal decentralization, the student should be able to empirically analyze the policy relevance of these tools.

UNIT -I

Modern Public Choice Theory and the Fiscal Decision process. Simple and Complex model of Majority Voting. Redistribution via the Fiscal Process, Stable growth as a Fiscal Objective, redefine Role of Government, fiscal Decision Making Process.

UNIT – II

The Budget Cycle and the Budgetary Process. Classification of Budgets and Budgeting Practices, Integration with National Accounts, Implementation, Authorization, Accountability and Audit, expenditure Structure and Policy Principle, Methods and Techniques of Expenditure Evaluation. Planning Programming Budgeting System, Zero- Base Budgeting System.

UNIT – III

Fiscal Policy, Meaning, Objectives, Short and Long Perspectives, Different Budget Adjustment with Investment, Fixed and Variable and in Closed and Open. Economic Growth and Fiscal Policy, recent and Contemporary Theoretical Foundations and Analytics, Applicability to Less Developed Countries.

UNIT – IV

Fiscal Decentralization, Finances of Local Governments, Financial Administration of Local Governments : Budget, Accounts and Audit.

Books Recommended:

Public Finance: Richard A. Musgrave **Public Finance**: Musgrave & Musgrave **Public Finance:** B.P.Tyagi

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV **Paper code: ECO-144**

PAPER IV: ADVANCED INDIAN ECONOMY -II

L	T	P	C
3	1	0	4

Course Objectives:

CO1: This course is designed to make students aware about the brief history, growth pattern and challenges of Indian Economy.

CO2: It will enable students to understand about the pattern of industrial growth in India as well as evolution of various economic policies in the industrial sector.

CO3: To understand the role of agriculture in economic development.Develop an understanding of the manifold obstacles to agricultural development, and the policies adopted to overcome them, with particular emphasis on the Indian agricultural scenario.

CO4: It will develop knowledge about the social sector and human development. It also delves into various social indexes.

Learning Outcome:

Understand the growth patterns of the Indian economy, industrial growth, policy shifts, major challenges, fiscal and monetary policies, and trade policies. To develop an idea about social sector development,human development, poverty, inequality, unemployment, women development, SHGs Brief History of Indian Economy; Major Challenges before Indian Economy.

UNIT I

Brief History of Indian Economy; Growth Pattern of Indian Economy; Sectoral Behavior and Relationship; Growth Pattern at Sub-National Level; Major Challenges before Indian Economy.

UNIT II

Pattern of Industrial Growth & Policy Shifts; Evolution of Economic Policies in India: Industrial Policy, Fiscal and Monetary Policies, Role of MSME sector in India, Exports – Import in industrial sector.

UNIT III

Pattern of Farm Sector Development, Agricultural Price Policy, Food Security, Modernisation & Diversification of the Farm Sector, WTO and Agriculture: Agreement on Agriculture (AoA), Direction of Export-Import in Agriculture, Other Activities of the Primary Sector, Natural Resources etc.

UNIT IV

Social Sector Development, Human Development, Poverty, Multidimensional Poverty Index, Inequality and its measures, Unemployment, Women Development, Gender Empowerment Index, Gender Inequality Index, Micro Finance & SHGs.

Books Recommended:

Indian Economy: Ruddar dutt & K.P.M. Sundaram. S. Chand
Indian Economy: A.N. Agarwal & M.K. Agarwal
Indian Economy: I.C. Dhingra
India after the Global Crisis: Shankar Acharya . Orient Blackswan
Indian Economy since Independence: Uma Kapila. Academic Foundation.

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV **Paper code: ECO-145A**

PAPER-V A-: FINANCIAL INSTITUTIONS AND MARKETS -II

L	T	P	C
3	1	0	4

Course Objectives:

CO1: This course will help students learn about role, structure and functioning of capital markets.

CO2: It will introduce students to the functioning of non bank financial intitutions ranging from mututal funds to pension funds and their regulatory framework in India.

CO3: It will develop in students knowledge about the discount and derivatives makets. It also delves into capital market and its working.

CO4: It will develop understanding among the students risk hedging and futures in exchange rates as well as reforms in international monetary system for developing countries.

Learning Outcome: After completing the course the students should be able to answer the basic questions regarding financial institutions, understand the working of the financial markets and the role of financial institutions in economic growth

UNIT – I

Role and Structure of Capital Market, Functions of investment banks, Primary and secondary capital markets, Securities underwriting and issuance, IPOs and bond offerings, Trading and brokerage operations.

UNIT – II

Definition and types of non-bank financial institutions: Insurance companies, Mutual funds, Pension funds, Role in the financial system, Regulatory framework for NBFIs, Risks and challenges in the non-banking sector, Impact of NBFIs on India's Economic development.

UNIT – III

Discount Market-Government Securities Market – Markets of Derivatives: Futures and Options, and Other Derivatives; Types, Uses and Pricing of Derivatives. SEBI: Its Impact on the Working of the Capital Market in India; IRDA and its Role in Financial Markets.

UNIT – IV

Risk Hedging and Futures in Exchange Rates, International Financial Flows, Forms and Volume – Rise and Fall of Brettonwood Institutions –International Monetary Fund (IMF) and World Bank, Reforms in International Monetary System for Developing Countries.

Books Recommended:-

Financial Market and Institutions: Mishkin. Pearson
Indian Financial System: Bharti Pathak McGraw Hill Education
Financial Institutions and Markets: L.M.Bhole & Jitendra Mahakud

CO-PO MAPPING:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV **Paper code: ECO-145B**

PAPER-V B- MATHEMATICS FOR ECONOMIC ANALSYSIS-II

L	T	P	C
3	1	0	4

Course objectives

CO1: This course is designed to help students learn about the concept and calculations of differential equations.

CO2: To help students learn about the concept and calculations of difference equations.

CO3: To learn the mathematical tools and concepts that aid in analyzing Input-Output Analysis and games theory.

CO4: It will develop knowledge and understanding of linear programming problems and various approaches to optimization.

Learning Outcome:

After completion of the paper the student should be able to understand basic mathematical tools that are used to analyze the economic data to arrive at the conclusion of theories. The tools include determinant geometry Differential calculus and integral calculus.

UNIT-I

Differential Equations ; Definitions and Concepts; First-order Linear Differential Equations; Exact
Differential Equations; Integrating Factors; Seperation of variables; Second-order Differential Equations. Economic Application.

UNIT-II

Difference Equations: Definitions and Concepts; First-order and Second-order, Linear Difference
Equations; Stability conditions. Economic Applications.

UNIT-III

Input-Output Analysis : Leonief's static open input-output model, Game theory- the zero- sum two-person game, pay-off matrices, Maximum and Minimax Strategies, Saddle point, Strategies.

UNIT-IV

Linear Programming : Graphic Approach; Simplex method; The Dual. Other approaches to optimization.

Books recommended:-

Mathematical Economics: Alfac. Chiang
Mathematics and Statistics for Economics: Mongia
Economics of Development and Planning: Mishra & Puri

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	1	1	1	1	2
CO2	3	2	2	1	1	2
CO3	3	3	3	1	1	2
CO4	3	1	1	1	1	2

3	**High Correlation**
2	**Moderate Correlation**
1	**Low Correlation**
0	**No Correlation**

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

M.A. ECONOMICS

SEMESTER – IV **Paper code: ECO-146**

PAPER VI-: MASTER DISSERTATION

L	T	P	C
1	0	2	4

Course Overview :

The objective of this course is to develop the research skills of the students of Master's Degree. The master dissertaion designed for students who want to engage in real economic research issues and analysis which will be much more than classroom teaching. The students will learn about the application of research tools and techniques in their own research work. The 04 credit project dissertation is designed to expose students to a variety of research problems in which researchers collect data and produce results on the basis of them which have high policy and practical implications.

General Outlines:

1. Each student should be assigned a project supervisor from the faculty members of the concerned center.
2. The topic of the project report should be decided by the supervisor taking into account the area of interest of the student. As far as possible the topic should address the current issues / developments in the field of Economics.
3. Standard norms for the preparation of a research project report may be followed – introduction, brief background, relevance of the topic, literature review, problem statement, objectives, hypotheses, sample design, data nature and sources, tools and techniques of data collection, data presentation, analysis, and interpretation, findings and conclusion.
4. The Dissertation outcomes may be reported in the following form:

Chapter-1: Introduction & Review of Literature;

Chapter-2: Research Methodology;

Chapter-3: Conceptual & Theoretical Description;

Chapter-4: Analysis and Interpretations;

Chapter-5: Conclusion, Suggestions & Scope for further research; References, Annexure, etc.

5. The Dissertation outcomes should be evaluated both in the form of reports and viva voce examination by a duly constituted committee by the appropriate authority.
6. This is to be evaluated as 'Satisfactory (S)' or 'Unsatisfactory (US)'